

Metabolic health is important for your overall well-being. From high triglycerides to high blood sugar, poor metabolic health is a risk factor for many chronic diseases. Check out the full articles for details, lifehacks, and references.

[Article: Diabetes](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MTNR1B	rs10830963	CC	G	0.27	Increased fasting glucose levels, increased risk of type 2 diabetes (2-fold) when eating late at night
TCF7L2	rs7903146	CC	T	0.28	Increased risk of diabetes, Decreased beta cell function.
TCF7L2	rs12255372	GG	T	0.28	Increased risk of diabetes due to decreased beta cell function.
SLC30A8	rs13266634	CT	T	0.28	T/T: lower risk for type-2 diabetes (zinc related)
IRS1	rs2943641	TC	C	0.65	Slightly higher risk for diabetes - common allele
WFS1	rs10010131	AA	A	0.37	Protective against diabetes
HHEX	rs7923837	GG	G	0.63	Increases risk of impaired glucose-stimulated insulin response
HHEX	rs1111875	CC	C	0.61	Increases risk of impaired glucose-stimulated insulin response
PPARG	rs1801282	CC	G	0.1	Increased risk of met. syndrome
KCNJ11	rs5219	TC	T	0.35	Decreased insulin response to glucose
CDKAL1	rs7754840	GG	C	0.4	increased risk for type 2 diabetes, decreased insulin release, impaired proinsulin to insulin conversion, increased diabetes risk even in people who are of normal weight
CDKAL1	rs7756992	AA	G	0.29	Increased relative risk of type 2 diabetes
HNF4A	rs2144908	--	A	0.18	Increased risk of diabetes in conjunction with other variants

[Article: Blood Glucose Regulation](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
KCNJ11	rs5219	TC	T	0.35	Impaired insulin secretion, higher blood glucose
ABCC8	rs757110	CA	A	0.65	AA: impaired insulin release
GCK	rs1799884	CC	T	0.17	Increased fasting plasma glucose
CDKAL1	rs7754840	GG	C	0.4	Decreased insulin release in response to glucose
IRS1	rs2943641	TC	C	0.65	CC: impaired insulin signaling
IRS1	rs1801278	CC	T	0.06	Impaired insulin signaling
ENPP1	rs1044498	AA	C	0.17	Increased insulin resistance
KCNH2	rs1805123	TT	G	0.22	Decreased fasting glucagon
GLP1R	rs6923761	--	A	0.3	Decreased insulin resistance, lower BMI (good!)
MTNR1B	rs10830963	CC	G	0.27	Increased fasting glucose if eating later at night

[Article: NAFLD](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PNPLA3	rs738409	CC	G	0.21	Significant increase in risk of NAFLD
TM6SF2	rs58542926	CC	T	0.06	Increased risk NAFLD
HSD17B13	rs6834314	AA	G	0.24	Liver fat, but less inflammation
MBOAT7	rs641738	CT	T	0.43	Increased risk NAFLD (Caucasian)
GCKR	rs1260326	CC	T	0.4	Increased triglycerides and fatty liver
CYP2E1	rs2031920	CC	T	0.02	Increased liver fat with alcohol
HFE	rs1800562	--	A	0.05	Iron build up, risk of NAFLD
HFE	rs1799945	CC	G	0.13	Iron build up, risk of NAFLD
PEMT	rs7946	CC	T	0.68	TT only: Increased NAFLD, choline related
SERPINA1	rs28929474	--	T	0.01	Alpha-1 antitrypsin, Increased risk of NAFLD and liver disease
SOD2	rs4880	AA	A	0.51	AA only: Increased fibrosis in NAFLD (oxidative stress)
IFNL3	rs12979860	TC	C	0.65	CC only: Increased liver damage in NAFLD
NCAN	rs2228603	--	T	0.06	Increased risk of NAFLD
AGTR1	rs5186	AA	C	0.27	increased risk of fatty liver, insulin resistance, and high blood pressure

[Article: PCOS](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
LHCRG	rs13405728	AG	G	0.08	Increased risk of PCOS
LHCRG	rs2293275	TC	T	0.41	Increased risk of PCOS (3-4x)
DENND1A	rs10818854	GG	A	0.04	Increased risk of PCOS (2x)
FSHB	rs11031006	--	A	0.13	Increased LH levels
FSHR	rs6166	TC	C	0.44	Increased risk of PCOS
ADIPOQ	rs2241766	TT	G	0.1	lower risk of PCOS
ADIPOQ	rs1501299	TG	T	0.26	Decreased risk of PCOS
MTNR1B	rs10830963	CC	G	0.27	Increased risk of PCOS - melatonin receptor
TNF	rs1799964	TC	C	0.21	increased TNF levels; increased relative risk of PCOS
TNF	rs4645843	--	T	0.002	increased relative risk of PCOS
INSR	rs2059807	GA	G	0.61	increased relative risk of PCOS
PLIN1	rs894160	TC	T	0.29	increased relative risk of obesity and diabetes in Asian populations, significantly increased relative risk of PCOS in women

[Article: GLP-1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GLP1R	rs10305492	GG	A	0.01	Lower fasting glucose
GLP1R	rs6923761	--	A	0.3	better response to GLP-1 receptor agonist for weight loss; DPP-4-inhibitors may not be as effective for diabetics
GLP1R	rs1042044	CC	A	0.44	AA: higher anhedonia
PCSK1	rs6232	TT	C	0.04	Increased risk of obesity; PCSK1 converts proglucagon into GLP1
PCSK1	rs6234	--	C	0.26	Increased risk of obesity
GCG	rs4664447	TT	C	0.006	CC: lower fasting glucose
GLP1R	rs10305420	--	T	0.33	better response to GLP-1 RAs for diabetes with greater decrease in HbA1c; increased relative risk of Alzheimer's
CNR1	rs1049353	CC	T	0.24	improvement in insulin resistance along with weight loss when using GLP-1 receptor agonist

[Article: Glucose and Caffeine](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ADORA2A	rs5751876	TT	T	0.44	T/T and C/T: no additive effect from caffeine plus carbs on blood glucose; CC: caffeine plus carbs increases postprandial glucose response
CYP1A2	rs762551	CA	C	0.31	Caffeine plus carbs causes glucose to remain elevated longer

[Article: Fasting](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GNB3	rs5443	CC	T	0.33	TT only: impacts mood and hunger when fasting (Hangry)

[Article: MCADD](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ACADM	rs77931234	AA	G	0.006	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	rs121434280	TT	C	0.0007	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	rs373715782	--	T	0.00005	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	i5012755	--	T	0.00003	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	rs121434281	--	T	0.00003	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	rs121434282	--	C	0	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	rs121434277	--	A	0.00003	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	rs121434274	GG	A	0.0001	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	i5012759	--	G	0.006	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	i5012760	--	T	0.005	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation
ACADM	i5003117	--	T	0.00003	Carrier of medium-chain acyl-CoA dehydrogenase deficiency mutation

[Article: SCADD](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ACADS	rs1800556	CC	T	0.03	Carrier of short-chain acyl-CoA dehydrogenase deficiency mutation
ACADS	rs28940874	--	T	0.0001	Carrier of short-chain acyl-CoA dehydrogenase deficiency mutation
ACADS	rs61732144	CC	T	0.001	Carrier of short-chain acyl-CoA dehydrogenase deficiency mutation
ACADS	rs28940872	--	T	0.0002	Carrier of short-chain acyl-CoA dehydrogenase deficiency mutation
ACADS	rs121908006	--	T	0.00009	Carrier of short-chain acyl-CoA dehydrogenase deficiency mutation
ACADS	i5007492	--	T	0.00009	Carrier of short-chain acyl-CoA dehydrogenase deficiency mutation
ACADS	i5007490	--	A	0	Carrier of short-chain acyl-CoA dehydrogenase deficiency mutation
ACADS	rs28941773	CC	T	0.0003	pathogenic for short-chain acyl-CoA dehydrogenase deficiency

[Article: SCD1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SCD	rs1393492	AA	G	0.08	Lower risk of met syndrome, lower SCD1
SCD	rs2060792	TT	T	0.71	TT only: higher palmitic and lower stearic acid, higher CRP
SCD	rs1502593	--	A	0.38	Increased risk of metabolic syndrome
SCD	rs7849	--	C	0.18	Lower BMI, increased insulin sensitivity
FTO	rs9939609	TT	A	0.39	Increased BMI, higher SCD1 with high carb
CYP1B1	rs1056836	GG	G	0.57	GG: Decreased estradiol metabolism

[Article: SIRTfoods](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SIRT1	rs12778366	TT	C	0.11	Reduced risk of diabetes complication, better glucose tolerance (good)
SIRT1	rs932658	--	A	0.44	Increased SIRT1 activity (usually good)
SIRT3	rs511744	--	T	0.3	T/T: increased lifespan
SIRT3	rs11246020	--	T	0.2	Reduced SIRT3; Increased risk of metabolic syndrome
SIRT3	rs185277566	--	G	0.04	Reduced SIRT3; Increased risk of heart attack
SIRT6	rs352493	TT	C	0.06	Increased severity in heart disease
SIRT1	rs3758391	--	T	0.34	Elevated SIRT1 expression, lower cardiovascular disease mortality (good)

[Article: Leptin](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
LEPR	rs1137101	AG	G	0.45	Increase risk for obesity; diabetes
LEPR	rs1137100	--	G	0.26	Increase risk for obesity; diabetes
LEPR	rs7799039	AG	A	0.28	Increase risk for obesity; diabetes
LEPR	rs3790433	--	T	0.27	Decreased risk for metabolic syndrome

[Article: MC4R; weight](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MC4R	rs17782313	CT	C	0.23	Increased BMI; Increased risk of obesity.
MC4R	rs17700633	AG	A	0.27	Increased BMI; Increased risk of obesity.
MC4R	rs12970134	AG	A	0.25	Increased BMI; Increased risk of obesity.
MC4R	rs571312	AC	A	0.23	Increased BMI; Increased risk of obesity.
MC4R	rs2229616	--	T	0.01	Significantly protective against obesity - rare

[Article: Brown Fat](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
UCP1	rs1800592	--	C	0.32	Increased risk for abdominal fat
UCP1	rs6536991	CT	C	0.24	Decreased risk of obesity
UCP1	rs3811787	--	G	0.29	Increased risk of abdominal fat

[Article: Adiponectin](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ADIPOQ	rs17300539	GG	A	0.07	Lower BMI; benefits from MUFA
ADIPOQ	rs1501299	TG	T	0.26	Higher adiponectin levels (may be diet specific)
ADIPOQ	rs266729	CC	G	0.09	GG only: better blood glucose with either carb rich or MUFA rich diet
ADIPOQ	rs2241766	TT	G	0.1	Increased risk of T2D

[Article: FTO](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FTO	rs9939609	TT	A	0.39	Increased risk of obesity; higher BMI
FTO	rs1558902	TT	A	0.38	Increased risk of obesity; higher BMI
FTO	rs3751812	GG	T	0.38	Increased risk of obesity; higher BMI

[Article: DHEA](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP2C9	rs2185570	TT	C	0.11	Lower DHEAS levels
TRIM4	rs17277546	GG	A	0.04	Lower DHEAS levels
SULT2A1	rs182420	TT	C	0.23	Somewhat Lower DHEAS levels
SULT2A1	rs2637125	GG	A	0.13	Lower DHEAS levels
SULT2A1	rs2910397	CC	T	0.3	Higher DHEAS levels

[Article: UCP2](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
UCP2	rs659366	CT	T	0.37	Higher BMI
UCP2	rs660339	GA	A	0.4	Higher BMI

[Article: GNB3](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GNB3	rs5443	CC	T	0.33	Enhanced G-protein signaling; Increased risk of obesity; diabetes

[Article: Ghrelin](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GHRL	rs4684677	TT	A	0.06	Increased ghrelin (hunger hormone) and BMI
GHRL	rs35683	--	A	0.46	Increased ghrelin (hunger hormone) and BMI
GHSR	rs572169	CC	T	0.29	Increased ghrelin (hunger hormone) and BMI
CLOCK	rs1801260	GA	G	0.26	Increased ghrelin (hunger hormone) and BMI
GHRL	rs696217	GG	T	0.07	Increased ghrelin (hunger hormone) and increased risk of obesity

[Article: Gout and Uric Acid](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ABCG2	rs2231142	GG	T	0.1	Increased risk of gout; higher uric acid
ABCG2	rs72552713	GG	A	0.0003	Increased risk of gout; higher uric acid
SLC2A9	rs6449213	CC	C	0.19	Decreased risk for gout; lower uric acid
SLC2A9	rs7442295	GG	G	0.22	Decreased risk for gout; lower uric acid
SLC2A9	rs12510549	--	C	0.2	Decreased risk for gout; lower uric acid
SLC2A9	rs12498742	--	G	0.29	Decreased risk for gout; lower uric acid
SLC2A9	rs16890979	--	T	0.23	Decreased risk for gout; lower uric acid
SLC2A9	rs1014290	--	G	0.25	Decreased risk for gout; lower uric acid
SLC2A9	rs10805346	CC	T	0.54	Decreased risk for gout; lower uric acid
SLC22A12	rs475688	--	C	0.74	Higher risk of gout (common)
GCKR	rs780094	CC	T	0.4	TT: higher risk of gout; CT and CC: lower risk of gout
SLC28A2	rs2271437	TT	G	0.002	Increased risk of gout

[Article: Adenovirus Weight Gain](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PCSK1	rs6232	TT	C	0.04	increased risk of infection by Ad-36, which is linked to weight gain
BDNF	rs4923461	AA	G	0.18	slightly increased risk of infection by Ad-36
PPARG	rs1801282	CC	G	0.1	increased risk of metabolic syndrome and insulin resistance, Ad-36 upregulates PPARG
CXADR	rs2824292	AG	G	0.46	reduced adenovirus 5 receptors (CXADR)
IL-10	rs1800871	AG	G	0.72	increase inflammation, higher BMI, and increased waist circumference
CXADR	rs437470	--	G	0.13	reduced adenovirus 5 receptors (CXADR)

Article: Carbohydrates

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IRS1	rs2943641	TC	C	0.65	C/C: no differences in diabetes risk based on low-carb vs low-fat diet (higher overall diabetes risk)
BDNF	rs6265	CC	T	0.18	Low-protein, high-carb (but not excessive calories) protective against T2D
FTO	rs9939609	TT	A	0.39	AA and AT: generally associated with higher BMI; TT: Greater decrease in HOMA-IR on lower fat diet
CLOCK	rs1801260	GA	G	0.26	less common variant, no benefit in insulin resistance for low-fat diet
UCP3	rs1800849	--	A	0.21	Less weight loss and no decrease in glucose or insulin levels on a low carb diet.

Article: FGF21 Sweet Tooth

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FGF21	rs838133	GG	A	0.42	Sweet preference, higher carb intake -- without increased weight

Article: Oxytocin and Weight Loss

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
OXTR	rs53576	--	A	0.32	higher fasting glucose and HOMA-IR; more at risk for obesity (adolescent study)

Article: Endocannabinoids and Weight

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MGLL	rs684358	--	G	0.03	linked with obesity (rare)
FAAH	rs324420	CC	A	0.2	reduced FAAH production leading to increased anandamide; higher BMI and waist circumference;
CNR1	rs806381	GG	G	0.32	Increased obesity, increased risk of PCOS, fatty liver disease
CNR1	rs12720071	--	C	0.1	Increased abdominal fat, increased risk of PCOS
CNR1	rs10485170	--	C	0.1	Increased risk of obesity, increased risk of PCOS
CNR1	rs1049353	CC	T	0.24	risk of increased visceral fat

Article: Diabetes and Coffee

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
C12orf51	rs2074356	GG	A	0.006	decrease in diabetes risk for coffee drinkers
ACAD10	rs11066015	GG	A	0.009	decrease in diabetes risk for coffee drinkers
MYL2	rs12229654	TT	G	0.007	decrease in diabetes risk for coffee drinkers

[Article: Melatonin levels](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PER3	rs228697	CC	G	0.1	may prefer to stay up late
MTNR1B	rs10830963	CC	G	0.27	linked to a higher risk of diabetes, increased fasting glucose – don't eat dinner late
AANAT	rs28936679	--	A	0.0005	may cause delayed phase sleep disorder

[Article: Hacking BDNF](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
BDNF	rs6265	CC	T	0.18	decreased BDNF, associated with an increased risk of introversion, suicide, and obesity (Met/Met)

[Article: Weight Loss Report](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FTO	rs9939609	TT	A	0.39	Increased BMI; increased risk of obesity (variant with the largest impact, circadian rhythm and appetite)
MC4R	rs17782313	CT	C	0.23	Increased BMI; increased muscle mass; higher appetite
MC4R	rs17700633	AG	A	0.27	Increased BMI; increased muscle mass; higher appetite
NPY	rs16139	TT	C	0.03	Increased BMI; higher leptin levels, increased food intake (hunger driven by brain)
LEPR	rs1137101	AG	G	0.45	Increased risk of being overweight or obese; weight gain on antipsychotic medications; greater muscle gain with weight training
LEPR	rs1137100	--	G	0.26	Increased risk of being overweight or obese; increased diabetes risk
SH2B1	rs7359397	CT	T	0.35	Higher BMI, increased risk of obesity, increased response to leptin
CLOCK	rs1801260	GA	G	0.26	Increased risk for obesity, harder to lose weight, evening chronotype, core circadian clock gene
SLC6A14	rs2011162	--	G	0.52	Decreased fatty acid oxidation; increased risk for obesity; serotonin synthesis gene
UCP1	rs1800592	--	C	0.32	Weak UCP1 activity, higher risk of abdominal fat, generally higher risk of obesity

[Article: GLP-1 RA weight loss](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GLP1R	rs6923761	--	A	0.3	better response to GLP1 receptor agonist for weight loss
GLP1R	rs10305492	GG	A	0.01	less response to GLP1 receptor agonist for weight loss
TCF7L2	rs7903146	--	T	0.28	less response to GLP1 receptor agonist for weight loss
CNR1	rs1049353	CC	T	0.24	improvement in insulin resistance along with weight loss in GLP-1 receptor agonist

[Article: FTO/m6A](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FTO	rs9939609	TT	A	0.39	Higher risk of obesity, increased BMI, increased FTO expression
FTO	rs8044769	TT	T	0.47	Increased BMI (~1.5 pts)
FTO	rs1558902	TT	A	0.38	Higher BMI, but not associated with obesity-related problems
FTO	rs3751812	GG	T	0.38	Higher BMI, no increase in diabetes risk
FTO	rs121918214	--	A	0	Rare mutation causing growth retardation, developmental delay, and early death
FTO	rs121918214	--	A	0	Rare mutation causing growth retardation, developmental delay, and early death

[Article: Microbiome and Weight](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
APOA5	rs651821	TT	C	0.07	Reduced Bifidobacterium levels, higher triglycerides, and MetS risk
MYD88	rs4988453	--	A	0.03	Reduced MYD88
MYD88	i5000725	--	C	0	Rare mutation, listed as pathogenic for MYD88 deficiency
MYD88	rs137853065	--	C	0	Rare mutation, listed as pathogenic for MYD88 deficiency
MYD88	i5000726	--	T	0.00003	Rare mutation, listed as pathogenic for MYD88 deficiency
MYD88	rs137853064	--	T	0.00003	Rare mutation, listed as pathogenic for MYD88 deficiency
NOD1	rs2075822	--	G	0.21	Increased risk of IBD
TLR4	rs10759932	--	C	0.11	Decreased risk of H. pylori
SLC39A8	rs13107325	CC	T	0.07	Changes in the gut microbiome, obesity, and Crohn's disease risk
TLR4	rs4986790	AA	G	0.05	Increased risk of gram-negative bacterial infection, septic shock, and metabolic syndrome

[Article: Diet Genes](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HNF1A	rs7957197	TT	T	0.81	Greater weight loss with a high-fat diet
MTNR1B	rs10830963	CC	G	0.27	Greater improvements in fat distribution, and weight loss with low-fat diet
IL-6	rs1800795	GG	C	0.34	CC only: Greater weight loss on a Mediterranean diet with additional olive oil
DHCR7	rs12785878	GG	T	0.66	Greater decreases in insulin and HOMA-IR in response to high-protein diets
ADCY3	rs10182181	GA	G	0.49	Less decrease in fat mass, greater lean mass loss on high-protein diet
PPM1K	rs1440581	CC	C	0.54	Less weight loss with high-fat diet
IRS1	rs2943641	TC	C	0.65	C/C only: more benefits in weight loss and improvement of insulin resistance on high-carbohydrate and low-fat diet
FTO	rs1558902	TT	A	0.38	high-protein diet worked best for weight loss
PLIN1	rs2289487	--	C	0.37	more weight loss with high protein, low-calorie diet
LIPF	rs814628	--	G	0.17	decreased gastric lipase, possibly more weight loss on a ketogenic diet
PPARG	rs1801282	CC	G	0.1	a higher risk of obesity with a high-fat diet
NPY	rs16147	TT	T	0.48	CT or TT: less weight loss on a high-fat diet, and better weight loss on a low-fat diet; C/C: better weight loss with a high fat diet
TCF7L2	rs12255372	GG	T	0.28	more weight-loss on a low-fat diet
AMY1	rs11185098	--	A	0.23	A/A: higher amylase activity, great at breaking down carbs, more likely to have higher BMI with high carbohydrate intake; A/G: intermediate amylase activity; G/G: lower amylase activity, less weight with a high-carb diet

[Article: HbA1c](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
G6PD	rs1050828	--	T	0.006	G6PD deficiency carrier; lower HbA1c levels
HBB	i3003137	--	A	0.0007	sickle cell trait carrier; lower HbA1c by 0.31% units
HFE	rs1800562	--	A	0.05	C282Y (hemochromatosis risk); often higher hemoglobin and lower HbA1c
TMPRSS6	rs855791	GA	A	0.47	lower ferritin levels; higher HbA1c levels (due to lower hemoglobin levels)
HK1	rs16926246	CC	T	0.12	slightly lower HbA1c levels
ANK1	rs4737009	GG	A	0.26	slightly higher HbA1c levels
MYO9B	rs11667918	CC	C	0.73	slightly higher HbA1c levels
FNK3	rs1046875	AG	A	0.33	slightly higher HbA1c levels
HBB	rs334	--	A	0.0007	sickle cell trait carrier; lower HbA1c by 0.31% units

[Article: Lipedema: Genes, Root Causes, Possible Solutions](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
POMC	rs202127120	--	A	0.0004	(rare) associated with lipedema risk
NR0B2	rs150160927	--	A	0.00004	(rare) associated with lipedema risk
RSPO3	rs72959041	GG	A	0.03	decreased association with lipedema
LHFPL6	rs1409440	TT	C	0.1	increased relative risk of lipedema
TMEM192	rs9308098	CT	C	0.12	increased relative risk of lipedema
SLC9B1P3	rs11511253	--	A	0.21	increased relative risk of lipedema

[Article: PPARG](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PPARG	rs1801282	CC	G	0.1	moderately reduced PPARG activity; generally lower risk of diabetes and lower weight gain on a normal diet; however, a higher risk of obesity with a high-fat diet
PPARG	rs10865710	CC	G	0.17	slightly higher glucose and lipid levels; increased risk of cardiovascular disease
PPARG	rs17036170	GG	A	0.01	increased risk of liver injury with diclofenac
PPARG	rs3856806	CC	T	0.12	decreased risk of type 2 diabetes
PPARGC1A	rs8192678	CC	T	0.32	decreased body fat

[Article: Snacking and Hunger](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FTO	rs9939609	TT	A	0.39	less likely to feel full, more hunger after a snack
LEPR	rs7799039	AG	A	0.28	less likely to feel full, more hunger after a snack
GNB3	rs5443	CC	T	0.33	bad mood and most hunger when fasting – hangry
FGF21	rs838133	GG	A	0.42	sweet preference, higher carb (lower protein and fat) intake without increased weight
LEPR	rs2025804	AA	G	0.32	G/G: increased snacking behavior, increased risk of obesity
TAS2R38	rs1726866	AA	A	0.52	A/A: More likely to have eating disinhibition in women
GAD2	rs992990	--	A	0.17	More likely to have emotional eating and eating disinhibition (women)
GAD2	rs7908975	--	C	0.14	More likely to have emotional eating and eating disinhibition (women)

[Article: Insulin Resistance](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PPM1K	rs1440581	CC	C	0.54	C/C: On average, higher BCAA levels compared to TT. For people on a reduced-calorie diet, a low-fat diet worked better for reducing insulin resistance.
PPM1K	rs9637599	--	C	0.5	On average, higher BCAA levels; Increased risk of insulin resistance and diabetes with high BCAA intake
BCAT1	rs2242400	--	G	0.11	increased BCAAs, increased risk of diabetes
GPRC6A	rs2274911	--	G	0.26	G/G: decreased risk of insulin resistance (amino acid receptor)
IRS1	rs1801278	CC	T	0.06	impaired IRS1 signaling, increased risk of insulin resistance and diabetes (insulin receptor)
ENPP1	rs1044498	AA	C	0.17	increased risk of insulin resistance, metabolic syndrome (helped by exercise))
PCK1	rs2179706	--	C	0.54	C/C: When consuming higher omega-3 PUFA, individuals had lower insulin resistance levels on average
IGF1	rs35767	GG	G	0.8	lower insulin sensitivity, increased risk of insulin resistance
NAT2	rs1208	GA	G	0.42	decreased risk of insulin resistance
IRS1	rs2943641	TC	C	0.65	C/C: increased risk of insulin resistance; C/T: typical risk; T/T: lower risk of type 2 diabetes in people with high vitamin D levels[

[Article: PLIN1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PLIN1	rs894160	TC	T	0.29	more likely to have increased insulin resistance with higher saturated fat intake; increased relative risk of PCOS
PLIN1	rs2289487	--	C	0.37	more weight loss with high protein, low-calorie diet, lower risk of obesity in men
PLIN1	rs6496589	--	G	0.01	increased risk of type 2 diabetes